ALGEBRA 1 · UNIT 10: FACTORING AND ZERO PRODUCT PROPERTY

Packaging Engineer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students work for Pack-It Packaging. They factor area and volume polynomials to find the dimensions of labels, box bases, and posters, solve quadratic equations with the Zero Product Property to meet size requirements, and design a box of their own.

Factoring is presented as finding dimensions from an area. The Zero Product Property then solves 'exact area' requests, and students must reject solutions that make a length negative or zero. A hands-on box from a sheet connects the algebra to a physical object.

Standards Addressed

- A-SSE.2 Use the structure of an expression to rewrite it (factoring)
- A-SSE.3a Factor a quadratic to reveal its zeros
- A-REI.4b Solve quadratic equations by factoring
- A-CED.1 Create equations and use them to solve problems

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: give a rectangle's area $x^2 + 7x + 12$ and ask for its sides. Part 1: factoring as dimensions.
Day 2	Part 2: exact-area requests and the Zero Product Property.
Day 3	Part 3: build the open box from a sheet; design and spec sheet.

Materials

- Student handout
- Grid paper and scissors (to build a box from a sheet)
- Rulers
- Calculators

How the Three Versions Differ

Level	What changes
CP2	GCF, $x^2 + bx + c$, perfect squares, and difference of squares as dimensions; ZPP with monic quadratics.
CP1	Adds $ax^2 + bx + c$, grouping, factoring completely for volume, and word problems that require rejecting a solution.
Honors	Factoring completely with several methods, cubic volume expressions factored by grouping, and a design with restrictions on x.

Watch For (Common Misconceptions)

- Setting a product equal to a nonzero number and using the ZPP (e.g., $(x + 3)(x + 4) = 30$ means $x + 3 = 30$).
- Keeping a negative length as an answer.
- Stopping before factoring completely ($2x^2 + 10x + 12 = (2x + 4)(x + 3)$).

Extension Ideas

- Find the cut size that gives the largest volume (table or graphing calculator).
- Price boxes by cardboard area.
- Design a cylinder label (preview of Geometry).

Answer Key – CP2

Part 1: Area to Dimensions

#	Task	Answer
1	A label has area $6x^2 + 9x$. Factor out the GCF to find its dimensions.	$3x(2x + 3)$: $3x$ by $(2x + 3)$
2	A box base has area $x^2 + 7x + 12$. Factor to find the dimensions.	$(x + 3)(x + 4)$
3	A square sticker has area $x^2 + 10x + 25$. What is the length of one side?	$(x + 5)^2 \rightarrow$ side $x + 5$
4	A square sheet x by x has a 4 by 4 square cut from its corner. The area left is $x^2 - 16$. Factor it to find the dimensions of a rectangle with the...	$(x + 4)(x - 4)$: a rectangle $(x + 4)$ by $(x - 4)$

Part 2: Exact Size Orders

#	Task	Answer
5	Solve $(x + 3)(x - 2) = 0$. Which solution could be a length? Explain.	$x = -3$ or $x = 2$. Only $x = 2$ can be used ($x = -3$ would make $x - 2$ negative and x itself negative).
6	A customer wants the base from #2, $(x + 3)(x + 4)$, to have an area of exactly 30 square inches. Find x and the base's dimensions.	$(x + 9)(x - 2) = 0 \rightarrow x = 2$ (reject -9). Dimensions 5 in by 6 in.
7	A square label has side $(x + 2)$ and area 49 square inches. Write and solve an equation for x.	$x^2 + 4x + 4 = 49 \rightarrow x^2 + 4x - 45 = 0 \rightarrow (x + 9)(x - 5) = 0 \rightarrow x = 5$ (label 7 in by 7 in)
8	A sticker has area $x^2 - 3x$ and must be 10 square inches. Find x.	$x^2 - 3x - 10 = 0 \rightarrow (x - 5)(x + 2) = 0 \rightarrow x = 5$

Part 3: Build a Box

#	Task	Answer
9	Write the length and width of the box's base in terms of x.	Length $10 - 2x$, width $8 - 2x$
10	The base must have an area of 48 square inches. Solve $(10 - 2x)(8 - 2x) = 48$. Which answer works? Why?	$80 - 36x + 4x^2 = 48 \rightarrow 4x^2 - 36x + 32 = 0 \rightarrow x^2 - 9x + 8 = 0 \rightarrow (x - 1)(x - 8) = 0$. $x = 1$ works; $x = 8$ gives negative sides.
11	Find the volume of your box. Then design a label for its front and write the label's area as a factored expression.	Base 8 in by 6 in, height 1 in: volume 48 cubic inches. Label designs vary.

Answer Key – CP1

Part 1: Area & Volume to Dimensions

#	Task	Answer
1	A box has volume $2x^3 + 10x^2 + 12x$. Factor completely to find three dimensions.	$2x(x^2 + 5x + 6) = 2x(x + 2)(x + 3)$
2	Labels have areas $2x^2 + 7x + 3$ and $3x^2 - 10x - 8$. Factor each.	$(2x + 1)(x + 3)$; $(3x + 2)(x - 4)$
3	A shipping crate has volume $x^3 + 3x^2 + 4x + 12$. Factor by grouping.	$x^2(x + 3) + 4(x + 3) = (x^2 + 4)(x + 3)$
4	Explain, using the cut-square picture, why $x^2 - 16 = (x + 4)(x - 4)$.	Cutting a 4 by 4 square from an x by x square leaves an L shape. Moving one strip makes an $(x + 4)$ by $(x - 4)$ rectangle.

Part 2: Exact Size Orders

#	Task	Answer
5	A label's length is 3 inches more than twice its width. Its area is 44 square inches. Find its dimensions.	$w(2w + 3) = 44 \rightarrow 2w^2 + 3w - 44 = 0 \rightarrow (2w + 11)(w - 4) = 0 \rightarrow w = 4$ in, length 11 in
6	A 20 in by 30 in poster gets a frame of width x all the way around. The framed poster has an area of 704 square inches. Find x.	$(20 + 2x)(30 + 2x) = 704 \rightarrow 4x^2 + 100x + 600 = 704 \rightarrow x^2 + 25x - 26 = 0 \rightarrow (x + 26)(x - 1) = 0 \rightarrow x = 1$ in
7	Cut squares of side x from a 10 in by 8 in sheet. The base must be 48 square inches. Find x and the volume.	$(x - 1)(x - 8) = 0 \rightarrow x = 1$. Box $8 \times 6 \times 1 \rightarrow 48$ cubic inches.
8	Solve $3x^2 - 10x = 8$ and decide which solution could be a length.	$3x^2 - 10x - 8 = 0 \rightarrow (3x + 2)(x - 4) = 0 \rightarrow x = 4$ or $x = -\frac{2}{3}$. Only $x = 4$ can be a length.

Part 3: Design a Box

#	Task	Answer
9	Choose your sheet size and a target base area that gives a whole-number cut size. Write and solve the equation.	Answers vary.
10	Find the box's dimensions and volume. Explain why one solution must be rejected.	Answers vary.
11	Write a spec sheet entry giving the base area in factored form and the volume as a factored polynomial in x.	Answers vary.

Answer Key – Honors

Part 1: Factor Completely

#	Task	Answer
1	$4x^3 - 36x$ and $x^4 - 16$	$4x(x^2 - 9) = 4x(x + 3)(x - 3)$ (GCF, difference of squares); $(x^2 + 4)(x^2 - 4) = (x^2 + 4)(x + 2)(x - 2)$ (difference of squares twice)
2	A crate has volume $V(x) = x^3 + 3x^2 - 4x - 12$. Factor it completely to find its dimensions.	$x^2(x + 3) - 4(x + 3) = (x^2 - 4)(x + 3) = (x + 3)(x + 2)(x - 2)$
3	For which values of x does the crate in #2 exist (all dimensions positive)? Find its volume when $x = 5$ two ways.	All factors positive when $x > 2$. $V(5) = 8 \cdot 7 \cdot 3 = 168$ and $125 + 75 - 20 - 12 = 168$ ✓
4	Show that $x^2 - 16 = (x + 4)(x - 4)$ using the cut-square picture, and use the same idea to explain why $x^2 + 16$ cannot be factored over the integers.	Removing a 4 by 4 square and rearranging gives $(x + 4)(x - 4)$. $x^2 + 16$ is a sum: nothing is removed, so no rectangle with integer-coefficient sides forms; no two integers multiply to 16 and add to 0.

Part 2: Exact Size Orders

#	Task	Answer
5	Squares of side x are cut from a 12 in by 12 in sheet. The base must be 64 square inches. Find x and the volume.	$(12 - 2x)^2 = 64 \rightarrow 4x^2 - 48x + 80 = 0 \rightarrow x^2 - 12x + 20 = 0 \rightarrow (x - 2)(x - 10) = 0 \rightarrow x = 2$ ($x = 10$ gives a negative side). Volume $8 \cdot 8 \cdot 2 = 128$ cubic inches.
6	A 20 in by 30 in poster gets a frame of width x. The frame itself (not the poster) has an area of 104 square inches. Find x.	$(20 + 2x)(30 + 2x) - 600 = 104 \rightarrow 4x^2 + 100x - 104 = 0 \rightarrow x^2 + 25x - 26 = 0 \rightarrow (x + 26)(x - 1) = 0 \rightarrow x = 1$ in
7	Solve $6x^2 + x = 15$. Which solution could be a length?	$6x^2 + x - 15 = 0 \rightarrow (3x + 5)(2x - 3) = 0 \rightarrow x = 1.5$ or $x = -1\frac{2}{3}$. Only 1.5 can be a length.
8	Find every x with $V(x) = 0$ for $V(x) = x^3 + 3x^2 - 4x - 12$. Explain what these values mean for the crate.	$x = -3, -2, 2$. A crate needs all positive dimensions, so none of these is a real crate (at $x = 2$ one side has length 0).

Part 3: Engineer a Package

#	Task	Answer
9	Write and solve the equation. Find the box's volume.	$(16 - 2x)(10 - 2x) = 72 \rightarrow 4x^2 - 52x + 160 = 72 \rightarrow x^2 - 13x + 22 = 0 \rightarrow (x - 2)(x - 11) = 0 \rightarrow x = 2$. Base 12×6, volume 144 cubic inches.
10	Explain the restriction on x (the values that give a real box) and use it to reject a solution.	$0 < x < 5$ (the width $10 - 2x$ must be positive), so $x = 11$ is rejected.

11	Design your own order: choose a sheet and a base area so that the cut size is a whole number. Write a spec sheet with the factored base area and…	Answers vary.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Box design meets the customer's area or volume exactly, and impossible solutions are rejected with a reason. Goes beyond the minimum with a creative, realistic choice.	Box design meets the customer's area or volume exactly, and impossible solutions are rejected with a reason.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Spec sheet shows the factored dimensions, the equation solved with the Zero Product Property, and a labeled net or drawing. Polished, organized, and visually engaging.	Spec sheet shows the factored dimensions, the equation solved with the Zero Product Property, and a labeled net or drawing. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.